

MGT 207

Production and Operations Management

Process Strategy and Analysis

Agenda

1. Process Strategies

- **Process Focus**
- **Repetitive Focus**
- **Product Focus**
- **Mass Customization**

2. Process Investment

- **Break-Even Analysis**
- Fixed vs. Variable Cost Structures
- Process Crossover Charts

3. Process Analysis

- **Capacity** (Design, Effective, Actual, Utilization, Efficiency)
- **Throughput & Bottlenecks**
- **Drum-Buffer-Rope (DBR)**

Process Strategy

The objective is to create a process to produce offerings that meet customer requirements within cost and other managerial constraints.

Customer Requirements

Quality, variety, delivery speed, and responsiveness.

Cost Constraints

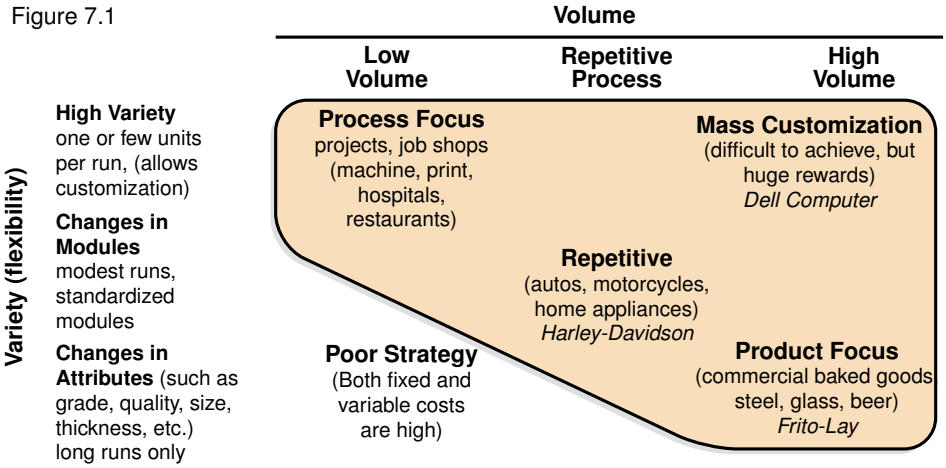
Capital investment (fixed cost) vs. unit variable cost.

Managerial Constraints

Capacity limits, technology, workforce, and flexibility.

Process, Volume, and Variety

Figure 7.1



Process Choice Based on Costs

Fixed vs. Variable Costs

Crossover Chart Example

- Evaluate three different accounting software products.
- Calculate crossover points between Software A and B and between Software B and C.

SOFTWARE	TOTAL FIXED COST	DOLLARS REQUIRED PER ACCOUNTING REPORT
Software A	$200,000	$60
Software B	$300,000	$25
Software C	$400,000	$10

Crossover Point V_1 (Software A vs. B)

$$\$200{,}000 + \$60\,V_1 = \$300{,}000 + \$25\,V_1$$

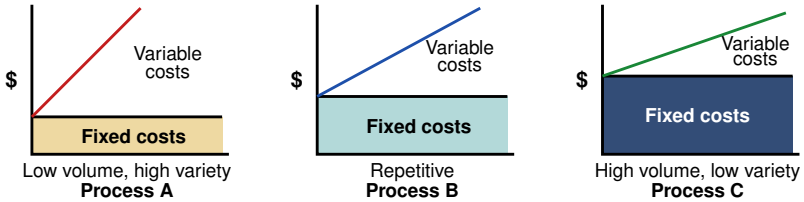
$$V_1 = \frac{\$100{,}000}{\$35} = \mathbf{2{,}857} \text{ reports}$$

Crossover Point V_2 (Software B vs. C)

$$\$300{,}000 + \$25\,V_2 = \$400{,}000 + \$10\,V_2$$

$$V_2 = \frac{\$100{,}000}{\$15} = \mathbf{6{,}667} \text{ reports}$$

Crossover Charts



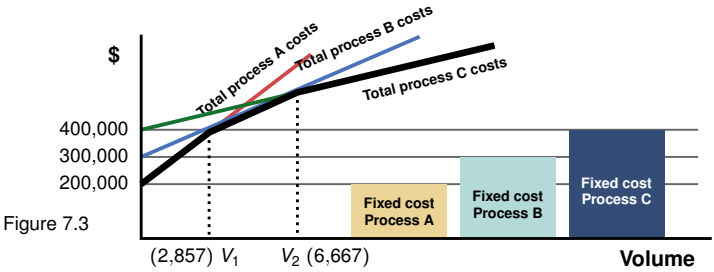


Figure 7.3

Process Choice Based On Capacity

Bottleneck Analysis

Outline

Part 1: Capacity

- Definition and Strategic Role of Capacity
- Design Capacity vs. Effective Capacity vs. Actual Output
- Utilization and Efficiency Metrics
- Managing Capacity and Demand

Part 2: Bottleneck Analysis & TOC

- Process Time, Flow Time, and System Capacity
- Identifying Bottlenecks (Single & Parallel Lines)
- Product-Mix Capacity Analysis (Keith's Car Wash)
- Theory of Constraints: Drum-Buffer-Rope (DBR)

Capacity

What is Capacity?

- The **throughput**, or the number of units a facility can hold, receive, store, or produce in a period of time.
- Determines **fixed costs**.
- Determines if **demand will be satisfied**.



Design and Effective Capacity

Design Capacity

- **Design capacity** is the **maximum theoretical output** of a system in a given period under ideal conditions.
- Normally expressed as a rate (e.g., bags/day, rolls/week, patients/hour).

Effective Capacity

- **Effective capacity** is the capacity a firm **expects to achieve** given current operating constraints.
- Often lower than design capacity due to product mix, maintenance, scheduling, and breaks.

Design Capacity $\geq$ **Effective Capacity** $\geq$ **Actual Output**

Design and Effective Capacity

TABLE S7.1 Capacity Measurements

MEASURE	DEFINITION	EXAMPLE
Design capacity 	Ideal conditions exist during the time that the system is available.	Machines at Frito-Lay are designed to produce 1,000 bags of chips/hr., and the plant operates 16 hrs./day. **Design Capacity** = 1,000 bags/hr. $\times$ 16 hrs. = **16,000** bags/day

Design and Effective Capacity

TABLE S7.1 Capacity Measurements

MEASURE	DEFINITION	EXAMPLE
Effective capacity 	Design capacity minus lost output because of *planned* resource unavailability (e.g., preventive maintenance, machine setups/changeovers, changes in product mix, scheduled breaks).	Frito-Lay loses 3 hours of output per day (= 0.5 hrs./day on preventive maintenance, 1 hr./day on employee breaks, and 1.5 hrs./day setting up machines for different products). **Effective Capacity** $= 16{,}000 - (1{,}000 \times 3)$ $= 16{,}000 - 3{,}000$ $=$ **13,000** bags/day

Design and Effective Capacity

TABLE S7.1 Capacity Measurements

MEASURE	DEFINITION	EXAMPLE
Actual output 	Effective capacity minus lost output during *unplanned* resource idleness (e.g., absenteeism, machine breakdowns, unavailable parts, quality problems).	On average, machines at Frito-Lay are not running 0.25 hr./day due to late parts and machine breakdowns. **Actual Output** $= 13{,}000 - (1{,}000 \times 0.25)$ $= 13{,}000 - 250$ $=$ **12,750** bags/day

Utilization and Efficiency

Utilization

Utilization is the percentage of **design capacity** actually achieved:

$$\textbf{Utilization} = \frac{\textbf{Actual Output}}{\textbf{Design Capacity}}$$

Efficiency

Efficiency is the percentage of **effective capacity** actually achieved:

$$\textbf{Efficiency} = \frac{\textbf{Actual Output}}{\textbf{Effective Capacity}}$$

Given Data (Single Line)

- Actual production last week = 148,000 rolls
- Effective capacity = 175,000 rolls
- **Design capacity** = 1,200 **rolls per hour**
- **Bakery operates** 7 **days/week, three** 8**-hour shifts**

Step 1: Compute Weekly Design Capacity

Multiply available operating hours per week by the hourly design rate:

$$\textbf{Design capacity} = (7 \times 3 \times 8) \times (1{,}200)$$
$$= \textbf{201,600} \text{ rolls/week}$$

7 days × 3 shifts × 8 hrs = 168 hrs/wk 1,200 rolls/hr

Given Data (Single Line)

- **Actual production last week** $= 148{,}000$ **rolls**
- Effective capacity $= 175{,}000$ rolls
- Design capacity $= 1{,}200$ rolls per hour
- Bakery operates 7 days/week, 3×8-hour shifts

Step 2: Compute Utilization

- **Design capacity:**

$$(7 \times 3 \times 8) \times 1{,}200 = \mathbf{201{,}600} \text{ rolls}$$

- **Utilization:**

$$\text{Utilization} = \frac{\mathbf{148{,}000}}{\mathbf{201{,}600}} = \mathbf{73.4}\%$$

Given Data (Single Line)

- **Actual production last week** $= 148{,}000$ **rolls**
- **Effective capacity** $= 175{,}000$ **rolls**
- Design capacity $= 1{,}200$ rolls per hour
- Bakery operates 7 days/week, 3×8-hour shifts

Step 3: Compute Efficiency

- **Design capacity:** $(7 \times 3 \times 8) \times 1{,}200 = 201{,}600$ rolls
- **Utilization:** $148{,}000/201{,}600 = 73.4\%$
- **Efficiency:**

$$\text{Efficiency} = \frac{\mathbf{148{,}000}}{\mathbf{175{,}000}} = \mathbf{84.6}\%$$

Bakery Example: Expansion

Given Data (Adding a Second Line)

- Actual production last week = 148,000 rolls
- Effective capacity = 175,000 rolls per line
- **Design capacity** = 201,600 **rolls per line**
- Efficiency = 84.6%
- Expected output of new line = 130,000 rolls

Expansion Step 1: Total Design Capacity

Adding a second identical production line doubles the design capacity:

Design capacity = **201,600** × 2 = **403,200** rolls

Bakery Example: Expansion

Given Data (Adding a Second Line)

- Actual production last week = 148,000 rolls
- **Effective capacity** = 175,000 **rolls per line**
- Design capacity = 201,600 rolls per line
- Efficiency = 84.6%
- Expected output of new line = 130,000 rolls

Expansion Step 2: Total Effective Capacity

- **Design capacity:**

$$201{,}600 \times 2 = 403{,}200 \text{ rolls}$$

- **Effective capacity:**

$$\mathbf{175{,}000} \times 2 = \mathbf{350{,}000} \text{ rolls}$$

Bakery Example: Expansion

Given Data (Adding a Second Line)

- **Actual production last week** $= 148{,}000$ **rolls**
- Effective capacity $= 175{,}000$ rolls per line
- Design capacity $= 201{,}600$ rolls per line
- Efficiency $= 84.6\%$
- **Expected output of new line** $= 130{,}000$ **rolls**

Expansion Step 3: Total Actual Output

- **Design capacity:** $201{,}600 \times 2 = 403{,}200$ rolls
- **Effective capacity:** $175{,}000 \times 2 = 350{,}000$ rolls
- **Actual output:**

$$148{,}000 + 130{,}000 = \mathbf{278{,}000} \text{ rolls}$$

Bakery Example: Expansion

Given Data (Two Lines)

- Line 1 actual output $= 148{,}000$ rolls
- Effective capacity $= 175{,}000$ rolls/line
- Design capacity $= 201{,}600$ rolls/line
- Line 1 efficiency $= 84.6\%$
- Expected output of new line $= 130{,}000$ rolls

Expansion Summary & New Ratios

- **Design capacity:** $201{,}600 \times 2 = \mathbf{403{,}200}$ rolls
- **Effective capacity:** $175{,}000 \times 2 = \mathbf{350{,}000}$ rolls
- **Actual output:** $148{,}000 + 130{,}000 = \mathbf{278{,}000}$ rolls

---

- **Utilization** $= \dfrac{278{,}000}{403{,}200} = \mathbf{68.95\%}$ (down from 73.4%)
- **Efficiency** $= \dfrac{278{,}000}{350{,}000} = \mathbf{79.43\%}$ (down from 84.6%)

Managing Capacity and Demand

1. Demand > Capacity

- Raise prices selectively
- Schedule longer lead times
- Discourage low-margin business
- **Long-term:** Expand capacity

2. Capacity > Demand

- Stimulate market demand
- Aggressive price promotions
- Introduce product changes
- Redevelop idle processes

3. Seasonal Demands

- Pair **complementary demand patterns**
- Counter-seasonal product lines
- Example: Snowmobiles (winter) + Jet skis (summer)

Process Analysis

1. Determine System Capacity and Throughput Time

- **Panda Express:** How many customers can a store serve per day?
- **Amazon:** How many packages can fulfillment centers process and ship per hour?

2. Identify and Manage Bottlenecks

- **Pinpoint Constraints:** Locate the stage with the lowest capacity limiting system output.
- **Debottleneck:** Reallocate resources, add parallel capacity, or buffer the bottleneck.

Capacity and Throughput

Process time	**Time** required to produce a unit at a workstation • Subways have three work stations
Throughput time (Flow time)	**Time** required to get through the entire process from start to end
Throughput	Average rate units actually being processed **(output/unit time)**
Capacity	Maximum average **output rate (output/unit time)** that the system can maintain under normal conditions

Throughput is Less Than or Equal to Capacity (Throughput $\leq$ Capacity)

What is a Bottleneck?

Bottleneck Definition

The step in the production/service delivery process:

a) that has the **lowest capacity** when measured **using a comparable unit of measure** and, therefore,

b) that limits the overall output of the system.

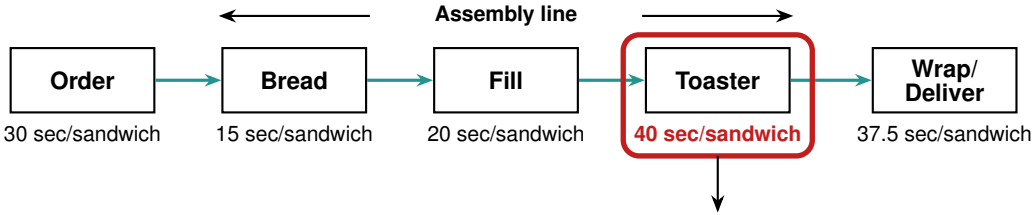


How Do You Find A Bottleneck?

a) On paper: Determine the capacity — **using a comparable measure** — of each step in the process.

b) Line of sight: Find the place in the production process that has the highest level of Work-In-Process (WIP) inventory.

Capacity Analysis

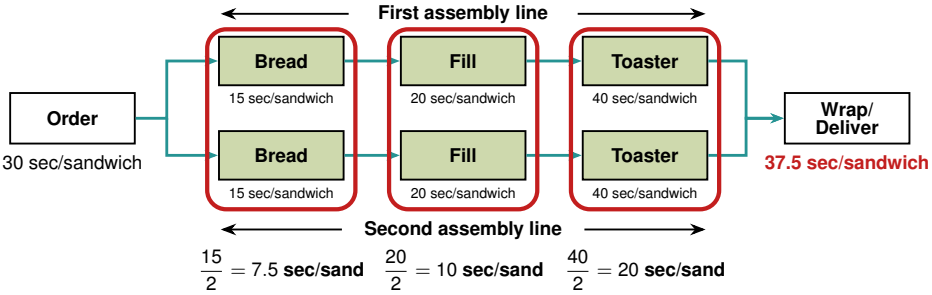


$$\text{Capacity} = 3600\ \frac{\text{sec}}{\text{hr}} \times \frac{1}{40}\ \frac{\text{sandwich}}{\text{sec}} = \mathbf{90}\ \text{sandwiches/hr}$$

$$\text{Throughput time} = 30 + 15 + 20 + 40 + 37.5 = \mathbf{142.5}\ \text{sec}$$

- Instead, we can convert processing times (sec/sandwich) to capacity of the individual stages (e.g. sandwiches/hr) and pick the stage with the lowest capacity.

Capacity Analysis: Parallel Processes



$$\text{Capacity} = \frac{1}{37.5}\ \frac{\text{sandwich}}{\text{sec}} \times 3600\ \frac{\text{sec}}{\text{hr}} = \mathbf{96}\ \text{sandwiches/hr}$$

$$\text{Throughput time} = 30+15+20+40+37.5 = \mathbf{142.5}\ \text{sec}$$

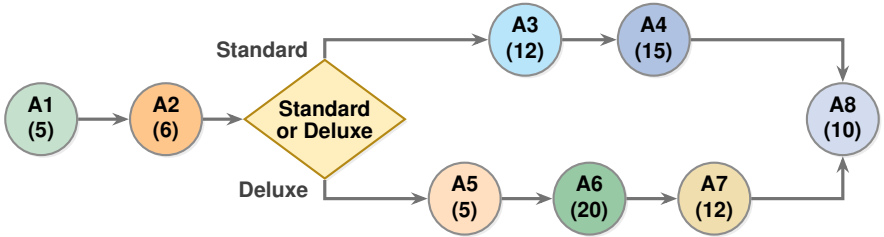
- **Has the capacity of the restaurant doubled after doubling the capacity of the (previous) bottleneck?** No—the bottleneck shifted to **Wrap/Deliver** (37.5 sec/sandwich), increasing capacity only from 90 to 96 sandwiches/hr.

Example 1 (1 of 4)

Keith's Car Wash: Process Structure

- **Two Offerings: Standard** wash vs. **Deluxe** wash *(numbers in parentheses = minutes per customer)*.
- **Shared Stages:** Both wash types enter through **A1** → **A2** and finish at drying station **A8**.
- **Dedicated Branches:** Standard goes through **A3** → **A4**; Deluxe goes through **A5** → **A6** → **A7**.

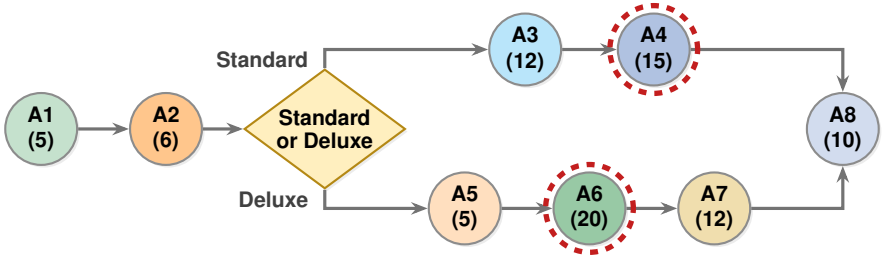
Figure 6.1 Process Flow for Keith's Car Wash



Q1

Q: Which step is the bottleneck for Standard vs. Deluxe washes?

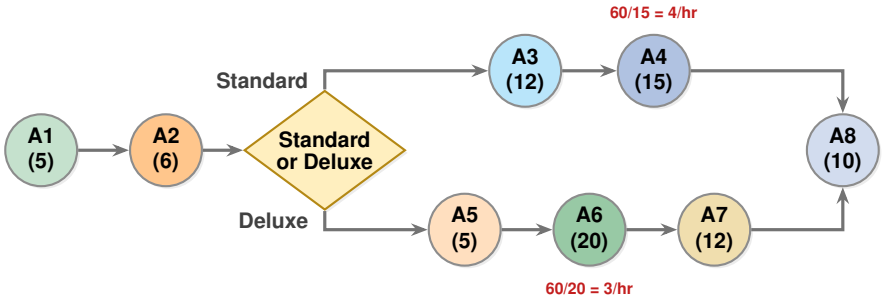
- **Standard Bottleneck:** Step **A4** (15 min/customer) — longest step on the Standard path.
- **Deluxe Bottleneck:** Step **A6** (20 min/customer) — longest step on the Deluxe path.



Q2

Q: What is the standalone capacity (customers/hr) for Standard and Deluxe?

- **Standard Capacity:** $\dfrac{60 \text{ min/hr}}{15 \text{ min (A4)}} = \mathbf{4}$ customers/hr *(assuming no wait at A1, A2, A8).*
- **Deluxe Capacity:** $\dfrac{60 \text{ min/hr}}{20 \text{ min (A6)}} = \mathbf{3}$ customers/hr *(assuming no wait at A1, A2, A8).*

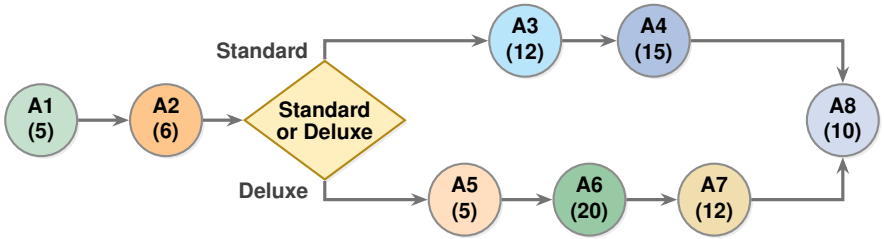


Q3

Q: With 60% Standard and 40% Deluxe, what is the average capacity?

A: Weighted average capacity across the customer mix:

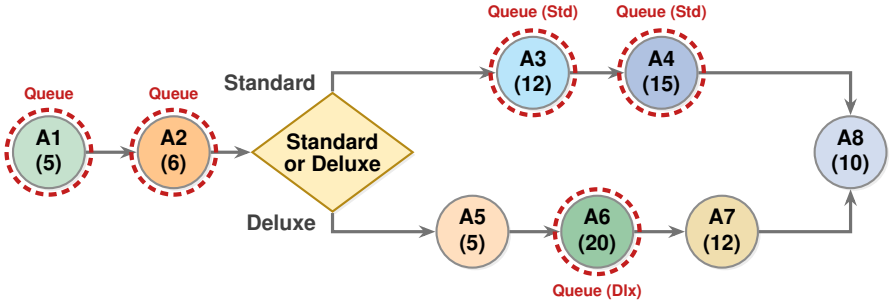
$$\text{Average Capacity} = (0.6 \times 4) + (0.4 \times 3) = 2.4 + 1.2 = \mathbf{3.6} \text{ customers per hour}$$



Q4

Q: Where do waiting lines form (continuous arrivals, one wash type at a time)?

- **Standard Queues:** Before **A1, A2, A3, and A4** (each step is slower than its predecessor: $5 < 6 < 12 < 15$ min).
- **Deluxe Queues:** Before **A1, A2, and A6** ($5 < 6$ and $5 < 20$ min; **A1** queues whenever arrivals exceed capacity).



Drum-Buffer-Rope Systems (1 of 3)

Drum-Buffer-Rope (DBR)

A planning and control system that regulates the flow of work-in-process (WIP) materials at the **bottleneck** or the **capacity constrained resource (CCR)** in a productive system.

The Drum

The bottleneck schedule that sets the beat (pace) for the entire plant.

The Buffer

Time buffer of WIP placed right before the bottleneck so it never starves.

The Rope

Communication signal tying material release to the bottleneck's consumption rate.

Drum-Buffer-Rope Systems (2 of 3)

1. The Drum (System Pace)

The **bottleneck schedule** is the **drum**—it sets the beat (production rate) for the entire plant and is linked directly to market demand.

2. The Buffer (Bottleneck Protection)

The **buffer** is a planned **time buffer** of WIP placed in front of the bottleneck to protect it from upstream disruptions.

3. The Rope (Material Release)

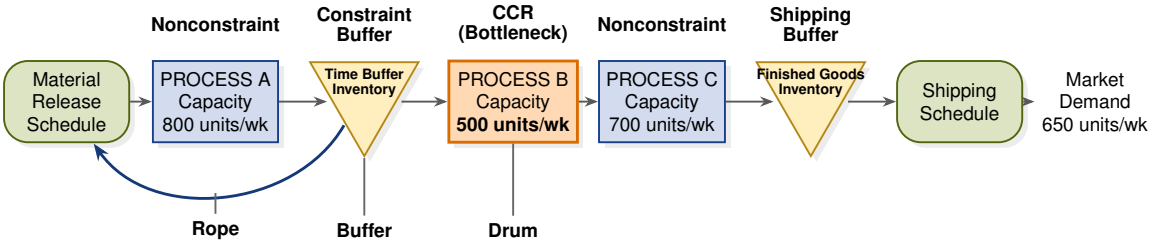
The **rope** ties upstream **material release** to the drumbeat, preventing excess WIP buildup ahead of the bottleneck.

4. Buffer Management (Execution)

Buffer management continuously monitors incoming bottleneck work to expedite late orders before the CCR starves.

Drum-Buffer-Rope Systems (3 of 3)

Figure 6.3 Drum-Buffer-Rope System



Summary

1. What a Bottleneck Is

- Point where work slows or stalls
- Narrowest part of a bottle
- Governs maximum system output rate

2. Why Bottlenecks Occur

- Machine breakdowns & setups
- Inefficient process steps
- Lack of skilled labor or parallel capacity

3. Identify the Bottleneck

- **On paper:** Lowest stage capacity (longest cycle time)
- **Line of sight:** Highest WIP inventory queue